Michael Brandenbursky

Curriculum Vitae — Updated 17 September 2026

PERSONAL DETAILS

E-mail brandens@bgu.ac.il

Webpage mbrandenbursky.github.io

ACADEMIC APPOINTMENTS

Associate Professor — November 2023–present
Ben-Gurion University, Beer Sheva, Israel

Senior Lecturer with tenure — May 2018–October 2023
Ben-Gurion University, Beer Sheva, Israel

Senior Lecturer — May 2017–April 2018
Ben-Gurion University, Beer Sheva, Israel

Lecturer — October 2015–April 2017
Ben-Gurion University, Beer Sheva, Israel

Researcher, CRM/ISM Fellowship — January 2014–October 2015
CRM, Montreal, Canada

Visiting Professor — May 2013–January 2014
Max Planck Institute for Mathematics, Bonn, Germany

Assistant Professor (NTT) — August 2010–August 2013
Vanderbilt University, Nashville, TN, USA

SHORT TERM RESEARCH POSITIONS

MFO, Germany	August 2025
ICMS, UK	July 2025
CRM, Canada	July–September 2024
University of Regensburg, Germany	July–September 2023
University of Regensburg, Germany	July–September 2022
CRM, Canada	July–September 2019
University of Regensburg, Germany	August–September 2018
University of Regensburg, Germany	July–August 2017
University of Regensburg, Germany	August–September 2016
MFO, Germany	June–July 2015
IHES, France	May–June 2015
MFO, Germany	June 2014
MFO, Germany	April 2013
MFO, Germany	June–August 2012
University of Aberdeen, UK	May–August 2011

RESEARCH VISITS

IMPA, Brazil	July–August 2026
University of Wrocław, Poland	May 2026
IMPA, Brazil	March 2025
IMPA, Brazil	April 2024
University of Pisa, Italy	April 2024
IMPA, Brazil	March 2023
IMPA, Brazil	August 2022
IMPA, Brazil	May 2022
IMPA, Brazil	March 2022
University of Aberdeen, UK	October 2021
University of Wrocław, Poland	September 2021
IMPA, Brazil	August 2019
University of Ceará, Brazil	June 2019
University of Wrocław, Poland	April 2019
University of Ceará, Brazil	February 2019
University of Aberdeen, UK	June 2018
University of Bochum, Germany	May 2018
University of Bonn, Germany	October 2017
IMPA, Brazil	September 2017
Universidad Nacional de Colombia, Medellín, Colombia	August 2016
University of Regensburg, Germany	February 2016
IMPA, Brazil	March 2015
Seoul National University, South Korea	October 2014
University of Nantes, France	November 2013
University of Aberdeen, UK	June 2013
University of Aberdeen, UK	May 2012
University of Chicago, USA	November 2011
University of Chicago, USA	April 2011
University of Chicago, USA	October 2009
MPI, Bonn, Germany	July 2008

EDUCATION

Ph.D. in Mathematics 2006–2010
Technion–Israel Institute of Technology, Haifa, Israel
Supervisors: Prof. Michael Entov and Prof. Michael Polyak. Thesis title: Knot invariants and their applications.

M.Sc. in Mathematics 2003–2005
Hebrew University, Jerusalem, Israel
Supervisor: Prof. Emmanuel Farjoun. Thesis title: Compact Lie groups.

B.A. in Mathematics & Economics 1999–2003
Hebrew University, Jerusalem, Israel

RESEARCH INTERESTS

Knot theory
Finite-type invariants, concordance group, knot Floer and Khovanov homologies

Symplectic geometry and low-dimensional topology
Groups of symplectic and Hamiltonian diffeomorphisms, Hamiltonian dynamics

Braid groups, mapping class groups, groups of diffeomorphisms
Entropy, quasi-morphisms, group cohomology, norms, metrics, distortion, orders

Geometric group theory
Quasi-isometric embeddings of groups, invariant word metrics

Algebraic geometry
Lipschitz geometry of singularities

GRANTS

ISF Individual Research Grant 823/23. Annual amount: 220,000 NIS (60,000 USD); total amount: 880,000 NIS (240,000 USD). Period: 2023–2027.

Co-PI, Leverhulme Trust Grant RPG-2017-159. Total amount: 232,793 GBP. Period: 2017–2021.

PI, GIF-Young Grant I-2419-304.6/2016. Total amount: 26,000 EUR. Period: 2017.

PI, ESF Grant #4824. Total amount: 2,000 EUR. Period: 2012.

ACADEMIC AWARDS

Simons Professorship, Canada — July 2024–September 2024
Humboldt research fellowship, Germany (60,000 EUR) — May 2021–October 2023
Simons CRM Scholar, Canada — July 2019–September 2019
Mittag-Leffler fellowship, Sweden — November 2015 (declined)
CRM-ISM fellowship, Canada (100,000 USD) — January 2014–October 2015
Oberwolfach Leibniz fellowship (twice), Germany, June–August 2012, April 2013, May 2014, June–July 2015 (30,000 USD)
Coleman-Cohen fellowship, UK (8,000 USD) — May–August 2011

OTHER AWARDS

Second prize, Moscow Mathematical Olympiad — 1989

PUBLICATIONS

MathSciNet (17 September 2026): 27 indexed publications; 182 citations in 94 citing publications.
103 unique citing authors; h-index 8 (calculated from the publication-level citation counts in the export).
Web of Science Core Collection (17 September 2026): 28 indexed publications; h-index 9.
200 citations (124 excluding self-citations); 93 citing articles (69 excluding self-citations).
Average 7.14 citations per publication. Per-article WoS counts use the report's Total column (including self-citations).
Google Scholar (17 September 2026): 363 citations; h-index 13; i10-index 14.
Since 2021: 211 citations; h-index 10; i10-index 10.
Journal metrics for all articles: JCR 2024, Mathematics (dataset updated 18 June 2025).
JIF: Journal Impact Factor; AIS: Article Influence Score; JR and Q: AIS-based rank and quartile.

1. **Michael Brandenbursky**[PI], Jarek Kędra[PI], Michał Marcinkowski[PI], and Egor Shelukhin[PI]. On quasi-morphisms and distortion in homeomorphism groups. *Commun. Contemp. Math.* Article 2650050, accepted/in press (2026).
   WoS citations: 0. JIF: 1.2; AIS: 1.019; JR: 75/474; Q1.

2. **Michael Brandenbursky**[PI] and Egor Shelukhin[PI]. The L^p-diameter of the space of contractible loops. *Bull. Lond. Math. Soc.* 58(5), e70396 (2026).
WoS citations: 0. JIF: 0.9; AIS: 1.064; JR: 71/474; Q1.
3. **Michael Brandenbursky**[PI] and Michał Marcinkowski[PI]. Volume and Euler classes in bounded cohomology of transformation groups. *Glasg. Math. J.* 67(1), 34–49 (2025).
WoS citations: 1. JIF: 0.4; AIS: 0.435; JR: 239/474; Q3.
4. **Michael Brandenbursky**[PI] and Misha Verbitsky[PI]. Non-commutative Barge-Ghys quasimorphisms. *Int. Math. Res. Not. IMRN* 2024(15), 11135–11158 (2024).
WoS citations: 1. JIF: 0.9; AIS: 1.177; JR: 55/474; Q1.
Erratum (2026): arXiv:2212.12958v3.
5. **Michael Brandenbursky** and Michael Khanevsky. C^0-gap between entropy-zero Hamiltonians and autonomous diffeomorphisms of surfaces. *Israel J. Math.* 255(1), 311–324 (2023).
WoS citations: 0. JIF: 0.8; AIS: 0.870; JR: 99/474; Q1.
6. Lev Birbrair, **Michael Brandenbursky**, and Andrei Gabrielov. Lipschitz geometry of surface germs in $\mathbb{R}^4$: metric knots. *Selecta Math. (N.S.)* 29(3), Paper No. 43, 20 pp. (2023).
WoS citations: 2. JIF: 1.2; AIS: 1.669; JR: 33/474; Q1.
7. **Michael Brandenbursky**, Michał Marcinkowski, and Egor Shelukhin. The Schwarz–Milnor lemma for braids and area-preserving diffeomorphisms. *Selecta Math. (N.S.)* 28(4), Paper No. 74, 20 pp. (2022).
WoS citations: 3. JIF: 1.2; AIS: 1.669; JR: 33/474; Q1.
8. **Michael Brandenbursky** and Michał Marcinkowski. Bounded cohomology of transformation groups. *Math. Ann.* 382(3–4), 1181–1197 (2022).
WoS citations: 6. JIF: 1.4; AIS: 1.562; JR: 37/474; Q1.
9. **Michael Brandenbursky** and Jarek Kędra. Fragmentation norm and relative quasimorphisms. *Proc. Amer. Math. Soc.* 150(10), 4519–4531 (2022).
WoS citations: 14. JIF: 0.8; AIS: 0.738; JR: 123/474; Q2.
10. **Michael Brandenbursky** and Egor Shelukhin. On the entropy norm on Ham(S^2). *Ann. Math. Québec* 45(1), 231–237 (2021).
WoS citations: 0. JIF: 0.4; AIS: 0.571; JR: 176/474; Q2.
11. **Michael Brandenbursky** and Arpan Kabiraj. On the entropy norm on the group of diffeomorphisms of closed oriented surface. *J. Topol. Anal.* 12(1), 105–111 (2020).
WoS citations: 0. JIF: 0.4; AIS: 0.528; JR: 191/474; Q2.
12. **Michael Brandenbursky** and Michał Marcinkowski. Aut-invariant norms and Aut-invariant quasimorphisms on free and surface groups. *Comment. Math. Helv.* 94(4), 661–687 (2019).
WoS citations: 8. JIF: 1.4; AIS: 1.625; JR: 34/474; Q1.
13. **Michael Brandenbursky** and Michał Marcinkowski. Entropy and quasimorphisms. *J. Mod. Dyn.* 15, 143–163 (2019).
WoS citations: 13. JIF: 1.0; AIS: 1.211; JR: 54/474; Q1.
14. **Michael Brandenbursky**, Jarek Kędra, and Egor Shelukhin. On the autonomous norm on the group of Hamiltonian diffeomorphisms of the torus. *Commun. Contemp. Math.* 20(2), 1750042, 27 pp. (2018).
WoS citations: 17. JIF: 1.2; AIS: 1.019; JR: 75/474; Q1.
15. **Michael Brandenbursky** and Egor Shelukhin. The L^p-diameter of the group of area-preserving diffeomorphisms of S^2. *Geom. Topol.* 21(6), 3785–3810 (2017).
WoS citations: 8. JIF: 2.2; AIS: 3.016; JR: 14/474; Q1.
16. **Michael Brandenbursky** and Jarek Kędra. The autonomous norm on Ham$(\mathbb{R}^{2n})$ is bounded. *Ann. Math. Québec* 41, 63–65 (2017).
WoS citations: 1. JIF: 0.4; AIS: 0.571; JR: 176/474; Q2.
17. **Michael Brandenbursky**. Concordance of certain 3-braids and Gauss diagrams. *Topology Appl.* 214, 180–185 (2016).
WoS citations: 1. JIF: 0.5; AIS: 0.277; JR: 342/474; Q3.
18. **Michael Brandenbursky**, Światosław R. Gal, Jarek Kędra, and Michał Marcinkowski. The cancellation norm and the geometry of bi-invariant word metrics. *Glasg. Math. J.* 58(1), 153–176 (2016).
WoS citations: 20. JIF: 0.4; AIS: 0.435; JR: 239/474; Q3.
19. **Michael Brandenbursky** and Egor Shelukhin. On the L^p-geometry of autonomous Hamiltonian diffeomorphisms of surfaces. *Math. Res. Lett.* 22(5), 1275–1294 (2015).
WoS citations: 10. JIF: 0.6; AIS: 1.001; JR: 78/474; Q1.

20. **Michael Brandenbursky** and Jarek Kędra. Concordance group and stable commutator length in braid groups. *Algebr. Geom. Topol.* 15(5), 2861–2886 (2015).
WoS citations: 6. JIF: 0.6; AIS: 0.776; JR: 117/474; Q1.
21. **Michael Brandenbursky**. Bi-invariant metrics and quasi-morphisms on groups of Hamiltonian diffeomorphisms of surfaces. *Internat. J. Math.* 26(9), 1550066, 29 pp. (2015).
WoS citations: 23. JIF: 0.7; AIS: 0.570; JR: 177/474; Q2.
22. **Michael Brandenbursky**. Link invariants via counting surfaces. *Geom. Dedicata* 173, 243–270 (2014).
WoS citations: 3. JIF: 0.5; AIS: 0.535; JR: 187/474; Q2.
23. **Michael Brandenbursky**. Coloring link diagrams and Conway-type polynomial of braids. *Topology Appl.* 161, 141–158 (2014).
WoS citations: 1. JIF: 0.5; AIS: 0.277; JR: 342/474; Q3.
24. **Michael Brandenbursky** and Jarek Kędra. Quasi-isometric embeddings into diffeomorphism groups. *Groups Geom. Dyn.* 7(3), 523–534 (2013).
WoS citations: 4. JIF: 0.8; AIS: 0.834; JR: 107/474; Q1.
25. **Michael Brandenbursky**. Invariants of closed braids via counting surfaces. *J. Knot Theory Ramifications* 22(3), 1350011, 21 pp. (2013).
WoS citations: 2. JIF: 0.4; AIS: 0.308; JR: 319/474; Q3.
26. **Michael Brandenbursky** and Jarek Kędra. On the autonomous metric on the group of area-preserving diffeomorphisms of the 2-disc. *Algebr. Geom. Topol.* 13(2), 795–816 (2013).
WoS citations: 26. JIF: 0.6; AIS: 0.776; JR: 117/474; Q1.
27. **Michael Brandenbursky**. Quasi-morphisms and L^p-metrics on groups of volume-preserving diffeomorphisms. *J. Topol. Anal.* 4(2), 255–270 (2012).
WoS citations: 10. JIF: 0.4; AIS: 0.528; JR: 191/474; Q2.
28. **Michael Brandenbursky**. On quasi-morphisms from knot and braid invariants. *J. Knot Theory Ramifications* 20(10), 1397–1417 (2011).
WoS citations: 20. JIF: 0.4; AIS: 0.308; JR: 319/474; Q3.

Papers in conference proceedings

1. **Michael Brandenbursky** and Egor Shelukhin. On the large scale geometry of the L^p-metric on the symplectomorphism group of the two-sphere. *IHES Preprints*, IHES/M/15/09.
2. **Michael Brandenbursky**. On the autonomous metric on groups of Hamiltonian diffeomorphisms of closed hyperbolic surfaces. *Oberwolfach Preprints*, OWP 2013-18.

Submitted

- **Michael Brandenbursky** and Misha Verbitsky. Barge-Ghys quasimorphisms taking values in semi-direct products. Preprint (2026). Author manuscript.
- **Michael Brandenbursky** and Lior Menashe. Bounded cohomology of measure-preserving homeomorphism groups of non-orientable surfaces. Submitted; arXiv:2506.02728 (2025; revised August 2026).

TEACHING EXPERIENCE

Instructor, Ben-Gurion University

Multi-variable calculus for CS (Undergraduate, Course coordinator), BGU	Fall 2025
Single-variable calculus for electrical engineering (Undergraduate, Course coordinator, taught jointly), BGU	Fall 2025
Multi-variable calculus for CS (Undergraduate, Course coordinator), BGU	Fall 2024
Topics in geometry and topology (Graduate)	Fall 2024
Single-variable calculus for electrical engineering (Undergraduate, Course coordinator, taught jointly)	Fall 2024
Multi-variable calculus for CS, BGU	Fall 2023

Single-variable calculus for electrical engineering, BGU	Fall 2023
Multi-variable calculus for CS, BGU	Spring 2023
Single-variable calculus for electrical engineering	Fall 2022
Multi-variable calculus for CS	Fall 2022
Single-variable calculus for CS, BGU	Fall 2021
Single-variable calculus for electrical engineering	Fall 2021
Single-variable calculus for CS	Summer 2021
Topics in geometry and topology (Graduate)	Fall 2020
Single-variable calculus for electrical engineering	Fall 2020
Single-variable calculus for electrical engineering	Fall 2019
Workshop for math majors	Fall 2019
Reading course on "Cohomology theories"	Spring 2019
Single-variable calculus for electrical engineering	Fall 2018
Workshop for math majors	Fall 2018
Topics in geometry and topology (Graduate)	Fall 2017
Single-variable calculus for electrical engineering	Fall 2017
Differential geometry	Spring 2017
Single-variable calculus for electrical engineering	Fall 2016
Linear algebra for mechanical engineering	Fall 2016
Multi-variable calculus for mechanical engineering	Spring 2016
Algebraic structures	Fall 2015

Instructor, McGill University

Linear algebra and geometry	Fall 2014
Multi-variable calculus 3	Fall 2014
Single-variable calculus 2	Fall and Summer 2014

Instructor, Vanderbilt University

Single-variable calculus	Fall 2011 and Spring 2012
Multi-variable calculus	Spring 2011 and Spring 2013
Advanced multi-variable calculus	Fall 2012

SUPERVISION OF POSTDOCTORAL FELLOWS

Dr. Aditya De Saha	2026–present
Dr. Priyadip Mondal	2024–present
Dr. Arpan Kabiraj	2017–2018
Dr. Michał Marcinkowski	2017

M.SC. STUDENTS

Assaf Tesler	2026–present
Amit Levinson-Sela	2023–2026
Lior Menashe	2023–2025
Rishi Kumar	2019–2020

LECTURE SERIES

Quasi-morphisms and bi-invariant metrics in algebra and geometry — 3 lectures, Sde Boker, Israel	February 2019

Entropy, metrics and quasi-morphisms — 3 lectures, Aberdeen University, UK	June 2018
Knots, braids and quasi-morphisms — 3 lectures, University of Regensburg, Germany	July 2017
Quasi-morphisms in algebra and geometry — 3 lectures, McGill University, Canada	March 2015
Quasi-morphisms and applications — 3 lectures, Seoul National University, South Korea	October 2014

TALKS IN CONFERENCES AND COLLOQUIUMS

Colloquium, Haifa University, Israel	June 2026
Colloquium, Bar-Ilan University, Israel	June 2026
Conference Metric Geometry of Singularities, Poland	May 2026
Geometria métrica e Topologia do Nordeste, Brazil	March 2025
Colloquium, Hebrew University, Israel	June 2024
Colloquium, Tel Aviv University, Israel	May 2024
Conservative Dynamics and Symplectic Geometry, Brazil	August 2023
Geometry and Singularities, Poland	July 2023
Colloquium, Technion, Israel	May 2023
DMV meeting, Germany	September 2022
36th Summer Topology Conference, Austria	July 2022
Workshop on Symplectic Dynamics, Portugal	June 2022
IMU meeting, Israel	July 2021
Spring Topology and Dynamics Conference, USA	May 2021
Simplicial volumes and bounded cohomology, Germany	September 2020
Manifolds and groups, Germany	February 2020
Conservative Dynamics and Symplectic Geometry, Brazil	August 2019
Cohomological methods, Poland	April 2019
IMU-PMS Meeting, Poland	September 2018
Bochum-Cologne-Munster seminar, Germany	May 2018
Colloquium, Hebrew University, Israel	November 2017
IMU Meeting, Israel	May 2017
Colloquium, Universidad Nacional de Colombia, Medellín	August 2016
Colloquium, Universidad de los Andes, Colombia	August 2016
Encuentro de Topologia, Geometria y Teoria de Representaciones, Colombia	July 2016
Joint International Meeting of MMS and IMU, Mexico	September 2015
Joint International Meeting of AMS and IMU, Israel	June 2014
Algebraic Structures in Low-dimensional Topology, Germany	May 2014
AMS Sectional Meeting, USA	March 2014
Colloquium, University of Hawaii, USA	February 2014
Colloquium, University of Iowa, USA	January 2014
Colloquium, University of Haifa, Israel	December 2013
Colloquium, Ben-Gurion University, Israel	December 2013
Colloquium, University of Cologne, Germany	November 2013
Colloquium, Göttingen University, Germany	November 2013
Colloquium, Groningen University, Netherlands	September 2013
Knots, Manifolds, and Group Actions, Poland	September 2013

Rolfsenfest-Low Dimensional Topology, Knots, and Orderable Groups, CIRM, France	July 2013
Colloquium, Heriot-Watt University, UK	May 2013
Mapping Class Groups and Categorification, Canada	April 2013
Colloquium, University of Manitoba, Canada	February 2013
Geometric Groups on the Gulf Coast, USA	April 2012
Spring Topology and Dynamics Conference, Mexico	March 2012
Knots in Washington, USA	December 2011
Research School and Workshop on Braids, Vietnam	January 2011
Knots in Washington, USA	December 2010
Geometric Group Theory, Hyperbolic Dynamics and Symplectic Geometry, Germany	July 2010
Summer school on link homology, France	July 2009

TALKS IN SEMINARS

Tel Aviv University, Israel	June 2026
University of Wrocław, Poland	May 2026
University of Pisa, Italy	April 2024
University of Regensburg, Germany	July 2022
Tel Aviv University, Israel	April 2022
IMPA, Brazil	March 2022
Aberdeen University, UK	October 2021
Wrocław University, Poland	September 2021
University of Regensburg, Germany	June 2021
University of Illinois at Chicago, USA	September 2019
University of Toronto, Canada	September 2019
McGill University, Canada	September 2019
UQAM, Canada	September 2019
University of Ceará, Brazil	June 2019
Wrocław University, Poland	April 2019
Tel Aviv University, Israel	March 2019
Aberdeen University, UK	June 2018
Tel Aviv University, Israel	May 2018
Technion, Israel	December 2017
BGU, Israel	November 2017
IMPA, Brazil	September 2017
University of Haifa, Israel	June 2017
University of Regensburg, Germany	September 2016
Tel Aviv University, Israel	April 2016
Technion, Israel	November 2015
Lille University, France	May 2015
MIT, USA	May 2015
Tulane University, USA	April 2015
IMPA, Brazil	March 2015
McGill University, Canada	November 2014
CAIST, South Korea	October 2014
Syracuse University, USA	April 2014

Université du Québec à Montréal, Canada, mini-course	March 2014
Université de Montréal	March 2014
McGill University, Canada	February 2014
MPIM, Bonn	January 2014
University of Nantes, France, mini-course	November 2013
Université Paris 7, France	November 2013
University of Cologne, Germany	October 2013
University of Virginia, USA	April 2013
University of Haifa, Israel	December 2012
Université du Québec à Montréal, Canada	November 2012
University of Illinois at Chicago, USA	October 2012
Vanderbilt University, USA	October 2012
University of Haifa, Israel	May 2012
Aberdeen University, UK	May 2012
Hebrew University, Israel	May 2012
University of Chicago, USA	November 2011
Vanderbilt University, USA, mini-course	October 2011
Vanderbilt University, USA	September 2011
University of Chicago, USA	April 2011
Vanderbilt University, USA	September 2010
University of Illinois at Chicago, USA	October 2009
University of Chicago, USA	October 2009
Caltech, USA	October 2009
University of Indiana at Bloomington, USA	October 2009
Columbia University, USA	October 2009
SUNY at Stony Brook, USA	September 2009
Technion, Israel, mini-course	July 2008

ORGANIZATION OF CONFERENCES

Workshop on Complex and Symplectic geometry, BGU	May 2018
Geometry and Topology section of the Israeli Mathematical Union annual meeting	June 2016

ACADEMIC SERVICES

Member of departmental prize committee	2024–2026
Member of BGU Senate	2022–2024
Member of departmental prize committee	2020–2021
Member of departmental hiring committee	2018–2019
Colloquium organizer at BGU	2017–present
Organization of Geometry and Topology section of the Israeli Mathematical Union annual meeting	June 2016
Member of Israeli Mathematical Union	2015–present
Ph.D. proposal evaluation of Mr. Mark Berezovik — Tel Aviv University	January 2026
Ph.D. proposal evaluation of Ms. Yara Hatoum — Technion–Israel Institute of Technology	October 2024
Ph.D. proposal evaluation of Mr. Adi Diktovsky-Tel Aviv University	May 2022
Ph.D. thesis evaluation of Mr. Roque Marqueza-doctoral student at Cinvestav, Mexico	November 2017
M.Sc. thesis evaluation of Mr. Leonid Vishnevsky-Tel Aviv University	December 2021

Referee for JEMS, Geometry & Topology, International Journal of Algebra and Computation, Journal of Topology and Analysis, Annales mathématiques du Québec, Journal of Modern Dynamics, International Journal of Mathematics, Forum Mathematicum, Journal of Symplectic Geometry, Communications in Contemporary Mathematics, Proceedings of AMS, Boletín de la Sociedad Matemática Mexicana, Proceedings of the Edinburgh Mathematical Society, Advances in Mathematics, 2012–present

MISCELLANEOUS

Languages: Russian (native), Hebrew (native), English (fluent), Spanish (fluent), Portuguese (basic), German (basic).